

BLUESTOCK MUTUAL FUND ANALYTICS CAPSTONE REPORT

1. EXECUTIVE SUMMARY

The Mutual Fund Analytics Capstone Project was designed as a comprehensive end-to-end data analytics system aimed at analysing the Indian mutual fund industry using real-world financial datasets. The primary objective of this project was to build a complete data pipeline that transforms raw, unstructured financial data into meaningful, actionable insights through a combination of data engineering, statistical analysis, and business intelligence visualization.

In this project, multiple datasets related to mutual funds were collected, cleaned, and integrated into a structured analytical system. These datasets included NAV history, fund master information, investor transaction records, SIP inflow data, portfolio holdings, and scheme performance metrics. Additionally, live NAV data was also fetched using an external API (mfapi.in) to simulate real-time financial data ingestion.

The project follows a complete data lifecycle approach consisting of data ingestion, data cleaning, transformation, storage in a relational database (SQLite), exploratory data analysis (EDA), advanced financial performance computation, and final visualization through a Power BI dashboard.

A key component of this project is the development of advanced financial metrics that help evaluate mutual fund performance from multiple perspectives. These include risk-adjusted return measures such as Sharpe Ratio and Sortino Ratio, market sensitivity indicators such as Beta and Alpha, and downside risk measures such as Value at Risk (VaR) and Conditional Value at Risk (CVaR). In addition, long-term performance indicators such as CAGR, rolling Sharpe ratios, and maximum drawdown were also implemented.

The final outcome of the project is an interactive Power BI dashboard that provides insights into industry trends, fund performance, investor behavior, and SIP market dynamics. This dashboard enables users to explore financial data interactively and make informed investment decisions.

Overall, this project demonstrates the practical application of data analytics in the financial domain and highlights how structured data pipelines can support better decision-making in mutual fund investments.

2. DATA SOURCES

The project integrates multiple structured and semi-structured datasets that collectively represent the mutual fund ecosystem. Each dataset plays a distinct role in the overall analysis pipeline and contributes to different stages of the analytics process.

2.1 Fund Master Data

The fund master dataset contains the core structural information about mutual fund schemes. This includes details such as scheme names, fund houses, categories, sub-categories, risk levels, expense ratios, and benchmark indices.

This dataset acts as the foundation for all analytical operations because it provides the necessary identifiers (such as AMFI scheme codes) required to link different datasets together. Without this dataset, it would not be possible to establish relationships between NAV data, transaction data, and performance metrics.

2.2 NAV History Data

The NAV history dataset contains daily Net Asset Value information for approximately 40 mutual fund schemes over a multi-year period from 2022 to 2026. NAV represents the per-unit market value of a mutual fund and is a critical metric for evaluating fund performance.

This dataset was used extensively for time-series analysis. It enabled the calculation of daily returns, volatility, rolling performance metrics, and drawdown analysis. Additionally, NAV trends were used to identify market cycles such as bull and bear phases.

2.3 Investor Transactions Data

The investor transaction dataset captures individual-level investment activity. It includes details such as SIP investments, lump sum contributions, redemption transactions, transaction dates, investor demographics, and geographic location.

This dataset is essential for understanding investor behavior patterns. It allows segmentation of investors based on investment habits, risk preferences, and consistency of investments over time.

2.4 SIP and Inflow Data

The SIP inflow dataset provides time-series information on systematic investment plans across different time periods. This dataset is particularly important for analyzing long-term investment behavior in the mutual fund industry.

It was used to identify trends in monthly inflows, seasonal investment patterns, and overall growth in SIP participation. It also helped in identifying peak investment periods such as the recorded high inflow of ₹31,002 crore in December 2025.

2.5 Portfolio Holdings Data

The portfolio holdings dataset contains sector-wise allocation details for mutual funds. It describes how different funds allocate their investments across sectors such as banking, IT, pharmaceuticals, and energy.

This dataset was primarily used for sector concentration analysis and calculation of the Herfindahl-Hirschman Index (HHI), which measures the level of diversification within a fund's portfolio.

2.6 Performance Metrics Dataset

This dataset includes pre-computed financial indicators such as Sharpe Ratio, Beta, Alpha, standard deviation, and maximum drawdown. These metrics provide a quick summary of fund performance and risk characteristics.

These values were further validated and expanded during the performance analysis phase of the project.

3. ETL DESIGN (EXTRACT – TRANSFORM – LOAD)

The ETL pipeline forms the backbone of the entire analytics system. It ensures that raw data from multiple sources is systematically processed and transformed into a clean, structured format suitable for analysis.

3.1 Extract Phase

During the extraction phase, data was collected from multiple sources including CSV files and external APIs. A total of 10 structured CSV datasets were used, along with live NAV data fetched from the mfapi.in API.

The extraction process involved loading each dataset into Python using Pandas. In addition, JSON responses from the API were parsed and converted into tabular format for consistency with existing datasets.

This phase ensured that all relevant financial data was available in a unified environment for further processing.

3.2 Transform Phase

The transformation phase involved extensive data cleaning and feature engineering. This step was critical to ensure data quality and analytical accuracy.

Missing values in NAV data were handled using forward-fill methods to maintain continuity in time-series data. Duplicate entries were removed to avoid inconsistencies in calculations. Transaction types were standardized into consistent categories such as SIP, Lump Sum, and Redemption.

Date fields were converted into proper datetime formats to enable time-series analysis. Additionally, NAV values were validated to ensure they were positive and realistic.

Beyond cleaning, several derived features were created. These include daily returns, compound annual growth rate (CAGR), Sharpe Ratio, Sortino Ratio, Beta, Alpha, and rolling performance indicators. These features significantly enhanced the analytical depth of the dataset.

3.3 Load Phase

In the final phase of the ETL pipeline, the cleaned and transformed data was loaded into a structured SQLite database named `bluestock_mf.db`.

A star schema design was implemented to optimize querying and analytical performance. The schema included dimension tables such as `dim_fund` and `dim_date`, as well as fact tables such as `fact_nav`, `fact_transactions`, `fact_performance`, and `fact_aum`.

This relational structure allowed efficient joins and enabled seamless integration with Power BI for visualization.

4. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis was conducted to understand underlying patterns in fund performance, investor behavior, and market trends.

4.1 Fund Performance Distribution

The analysis revealed that large-cap mutual funds generally exhibited lower volatility and more stable returns compared to other categories. Small-cap funds, on the other hand, showed higher volatility but also offered higher potential returns. Index funds demonstrated consistent performance that closely tracked market indices.

4.2 NAV Trend Analysis

NAV trend analysis showed a clear upward movement across most funds during the period from 2022 to 2025. The year 2023 was characterized by a strong bull market, while 2024 experienced moderate corrections. These fluctuations highlight the cyclical nature of financial markets.

4.3 SIP Inflow Trends

SIP inflows showed a consistent upward trajectory over time, indicating growing investor participation in systematic investment plans. The highest recorded inflow was ₹31,002 crore in December 2025, suggesting strong investor confidence during that period.

4.4 Investor Behavior Analysis

Investor behavior analysis revealed that SIP investments were significantly more popular than lump sum investments. Redemption activity tended to increase during periods of market volatility, indicating reactive investor behavior.

Additionally, increasing participation from Tier-2 and Tier-3 cities was observed, suggesting improved financial inclusion across India.

4.5 Geographic Distribution

Geographic analysis showed that urban regions contributed the majority of SIP inflows. However, smaller cities are increasingly participating in mutual fund investments, indicating a gradual expansion of financial awareness.

4.6 Correlation Analysis

Correlation analysis between funds revealed that large-cap funds tend to move together due to similar underlying market exposure. In contrast, debt and equity funds showed weaker correlation, indicating diversification benefits for investors.

4.7 Sector Allocation Analysis

Sector allocation analysis revealed that financial services and information technology sectors dominate mutual fund portfolios. However, the degree of sector concentration varied significantly across funds, impacting overall risk exposure.

5. PERFORMANCE ANALYSIS

This section focuses on evaluating mutual fund performance using advanced financial metrics and statistical models.

5.1 Daily Returns Analysis

Daily returns across all mutual fund schemes were observed to be centered around zero, with limited extreme outliers. This indicates that the dataset reflects realistic market behavior and does not contain abnormal price distortions.

5.2 CAGR Analysis

Compound Annual Growth Rate (CAGR) analysis showed significant variation across funds, indicating differences in long-term fund management effectiveness. Some funds demonstrated strong compounding ability, while others showed relatively moderate growth.

5.3 Sharpe Ratio Analysis

The Sharpe Ratio analysis revealed that funds with higher Sharpe values consistently delivered better risk-adjusted returns. This confirms that evaluating returns alone is insufficient without considering associated risk.

5.4 Sortino Ratio Analysis

The Sortino Ratio analysis provided deeper insight into downside risk. It showed that certain funds were more efficient in minimizing negative returns while maintaining growth potential.

5.5 Alpha and Beta Analysis

Alpha analysis indicated that several funds generated positive excess returns over benchmark indices, suggesting effective active management. Beta analysis showed varying sensitivity of funds to market movements, reflecting different risk profiles.

5.6 Maximum Drawdown Analysis

Maximum drawdown analysis identified the worst historical decline for each fund. This metric is crucial for understanding investor risk exposure during extreme market conditions.

5.7 Fund Scorecard Model

A composite fund scorecard was developed to rank funds based on multiple performance factors. The scoring model included return performance, Sharpe ratio, Alpha, expense ratio, and maximum drawdown.

This multi-factor approach ensures a balanced evaluation rather than relying on a single metric.

5.8 Benchmark Comparison

Top-performing funds were compared against benchmark indices such as Nifty 50 and Nifty 100. Several funds either outperformed or closely tracked the benchmarks, while others showed higher tracking errors.

5.9 Key Performance Insights

The following key insights were observed:

- Returns were generally stable and realistic across funds.
- CAGR varied significantly across schemes.
- Sharpe Ratio strongly correlated with fund quality.
- Sortino Ratio highlighted downside protection efficiency.
- Alpha indicated active management value addition.
- Beta showed varying market sensitivity.
- Scorecard enabled balanced ranking.
- Top funds performed close to or above benchmarks.
- Tracking error varied significantly across funds.
- Multi-metric evaluation is essential for accurate analysis.

6. INVESTOR ANALYSIS

Investor analysis focused on understanding behavior patterns, investment consistency, and segmentation.

6.1 Cohort Analysis

Investors were grouped based on their first investment year. The 2024 cohort showed relatively lower investment amounts, while the 2025 cohort demonstrated higher engagement and stronger participation in SIP investments.

6.2 SIP Continuity Analysis

SIP continuity analysis revealed that many investors did not maintain consistent investment patterns. Investors with gaps exceeding 35 days were categorized as “at-risk,” indicating potential discontinuation of systematic investments.

6.3 Behavioral Insights

Overall behavioral analysis showed that SIP is the dominant investment strategy among retail investors. Younger investors tend to prefer high-risk funds, while more mature investors prefer stable, large-cap funds.

7. RECOMMENDATION SYSTEM

A rule-based recommendation system was developed to assist investors in selecting suitable mutual funds based on their risk appetite.

The system categorizes funds into three main groups:

- Low Risk: Debt funds and stable large-cap funds
- Moderate Risk: Balanced hybrid funds
- High Risk: Small-cap and aggressive equity funds

Funds are ranked based on Sharpe Ratio, historical performance, and risk classification to ensure accurate matching with investor profiles.

8. LIMITATIONS

Although the project provides comprehensive insights, certain limitations exist. The dataset is historical in nature and does not guarantee future performance. External macroeconomic variables such as inflation and interest rate changes were not incorporated into the analysis.

Additionally, investor behavior was analyzed purely based on transactional data, without considering psychological or external financial influences. Real-time market fluctuations were also not fully integrated into the model.

9. CONCLUSION

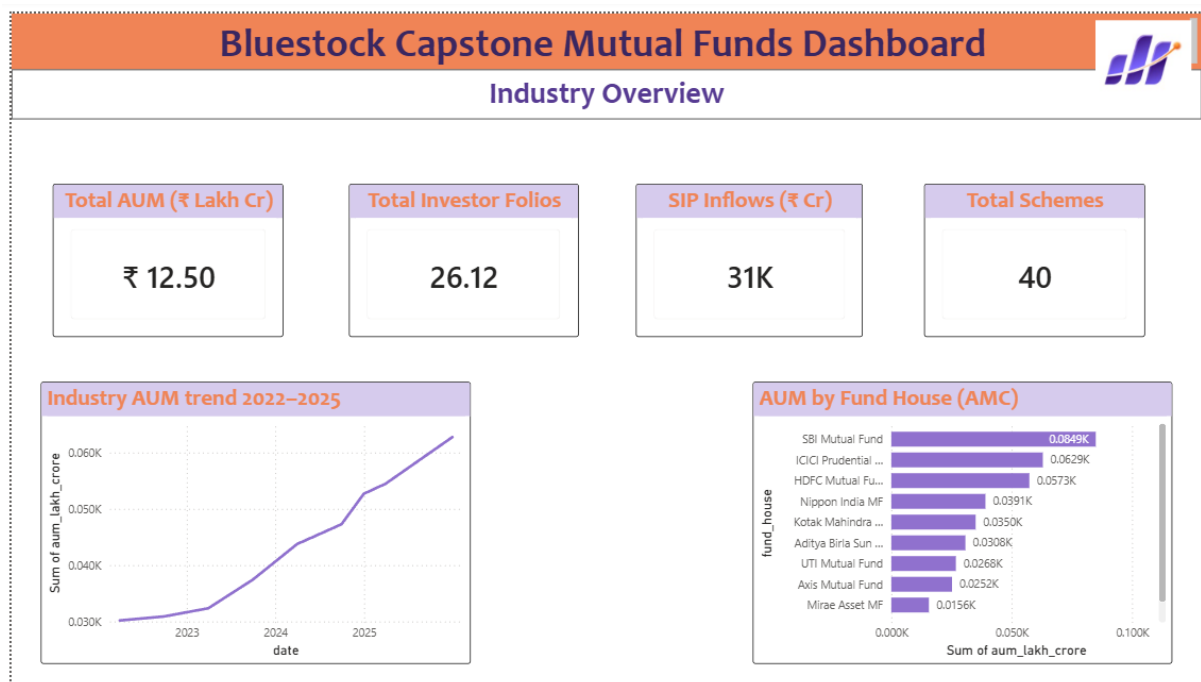
The Bluestock Mutual Fund Analytics Capstone Project successfully demonstrates a complete data analytics pipeline covering data engineering, financial modeling, investor behavior analysis, and business intelligence visualization.

The project provides meaningful insights into mutual fund performance, investor behavior, and market trends. By integrating multiple analytical techniques, the system enables a holistic understanding of financial markets.

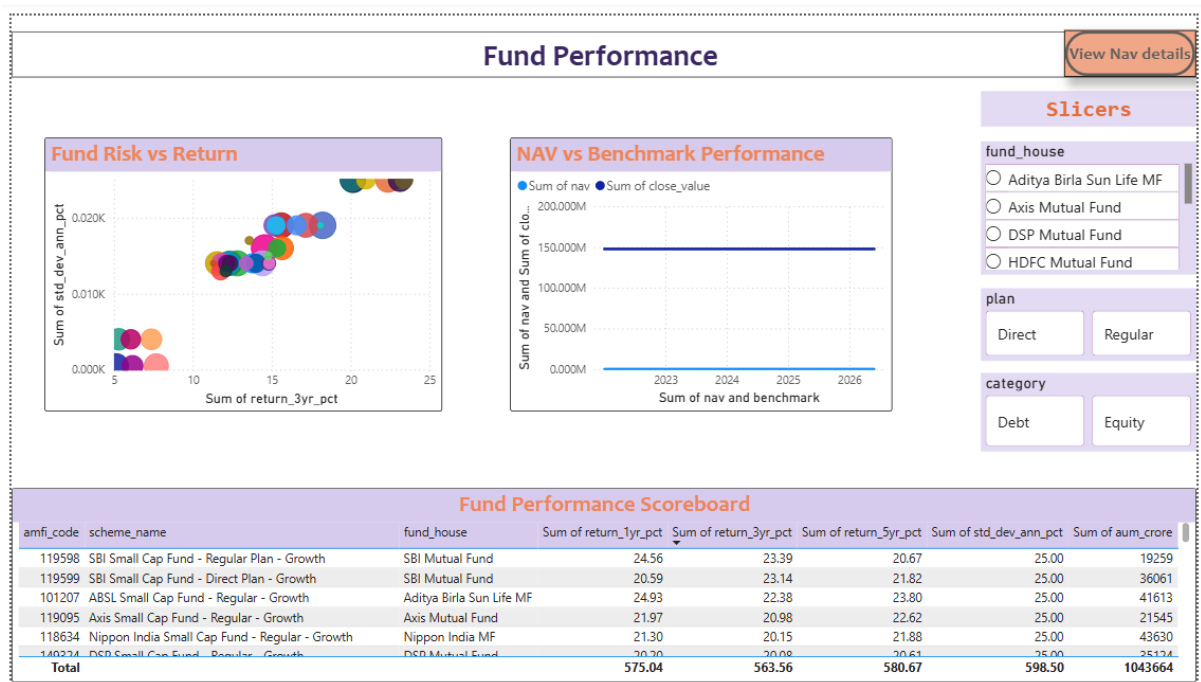
This project highlights the importance of combining data engineering with financial analytics to support informed investment decision-making. It also demonstrates how structured data pipelines and advanced statistical models can be used to derive actionable insights from complex financial datasets.

Overall, the project serves as a strong example of real-world application of data analytics in the financial domain.

Industry Overview



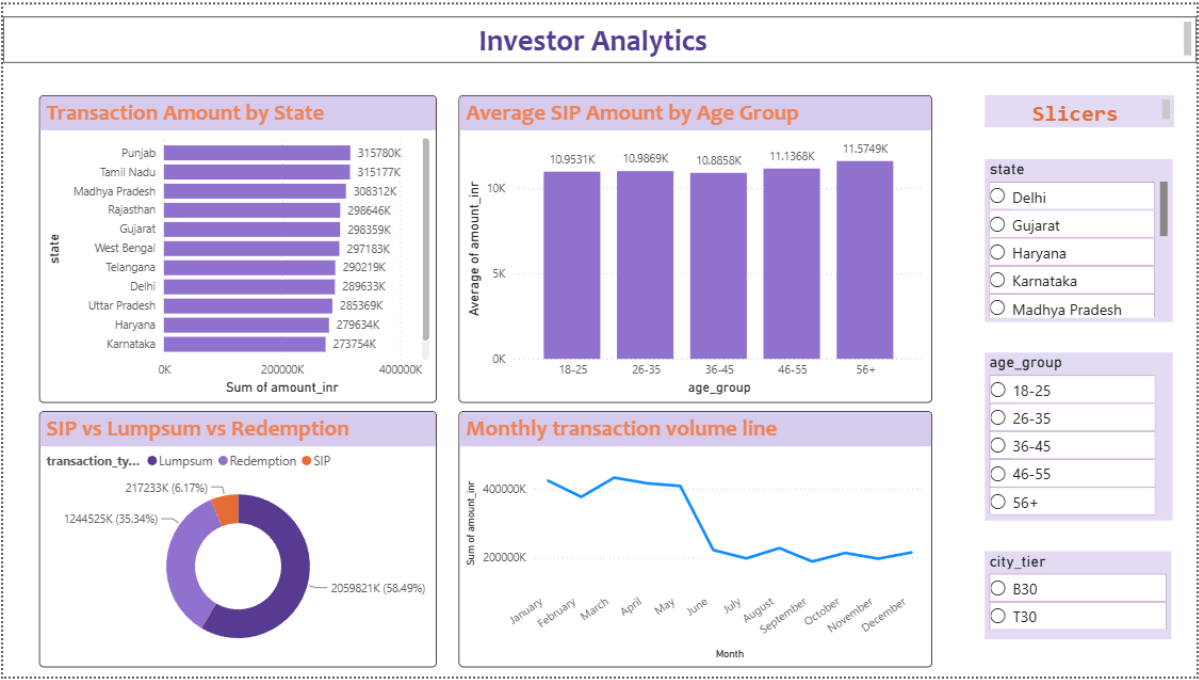
Fund Performance



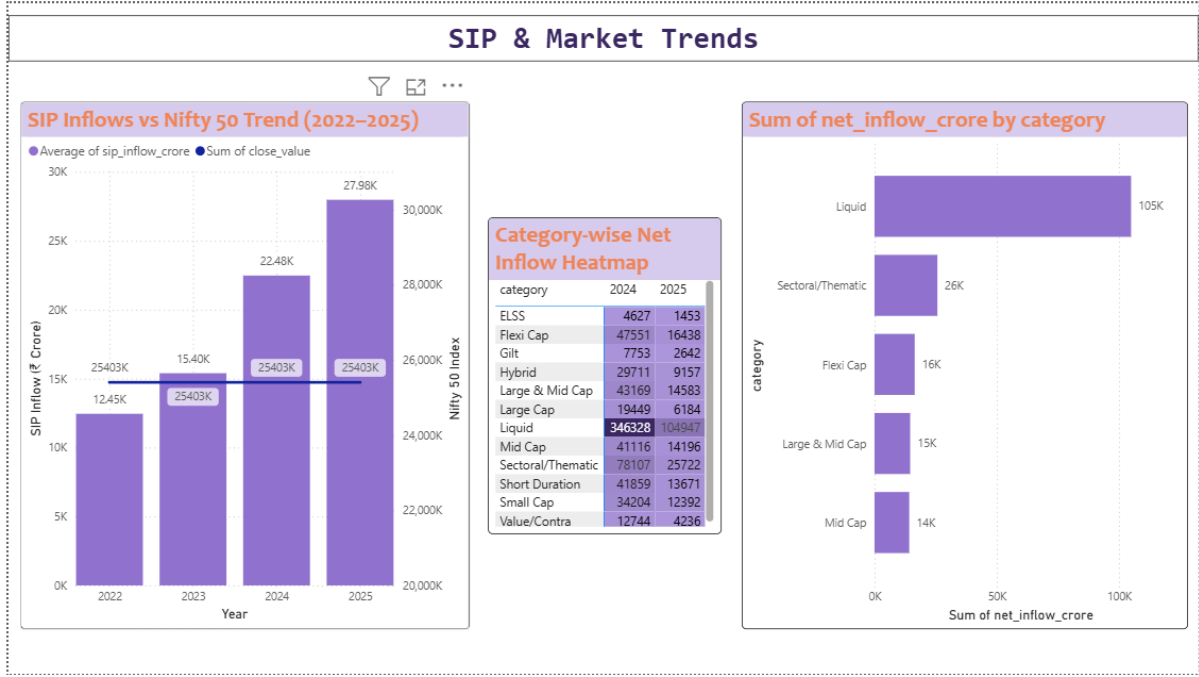
Fund Performance Scoreboard

amfi_code	scheme_name	fund_house	Sum of return_1yr_pct	Sum of return_3yr_pct	Sum of return_5yr_pct	Sum of std_dev_ann_pct	Sum of aum_crore
119598	SBI Small Cap Fund - Regular Plan - Growth	SBI Mutual Fund	24.56	23.39	20.67	25.00	19259
119599	SBI Small Cap Fund - Direct Plan - Growth	SBI Mutual Fund	20.59	23.14	21.82	25.00	36061
101207	ABSL Small Cap Fund - Regular - Growth	Aditya Birla Sun Life MF	24.93	22.38	23.80	25.00	41613
119095	Axis Small Cap Fund - Regular - Growth	Axis Mutual Fund	21.97	20.98	22.62	25.00	21545
118634	Nippon India Small Cap Fund - Regular - Growth	Nippon India MF	21.30	20.15	21.88	25.00	43630
140224	DSP Small Cap Fund - Regular - Growth	DSP Mutual Fund	20.20	20.08	20.61	25.00	26124
Total			575.04	563.56	580.67	598.50	1043664

Investor Analysis



SIP Market Trends



All pages include slicers for dynamic filtering and interactive exploration.